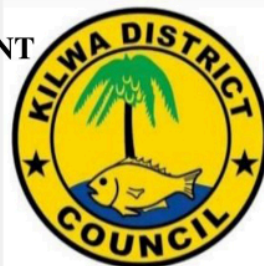




THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
KILWA DISTRICT COUNCIL
FORM THREE TERMINAL EXAMINATION



CHEMISTRY

GEPAM science hub

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Time: 3 Hours

May, 2026

INSTRUCTIONS

1. This paper consists of section **A, B** and **C** with total number of eleven (11) questions.
2. Answer **all** questions in section A and B and any **two (2)** questions in the section C
3. Section A carries **sixteen (16)** marks, section B carries **fifty-four(54)** marks and section C carries **thirty (30)** marks
4. All writing must be in **black** or **blue** pen except for diagrams which must be in **pencil**.
5. Non-programmable calculators may be used
6. Cellular phones and other unauthorized materials are not allowed in the examination room
7. The following constants may be used; $1F = 96500C$ $GMV = 22.4dm^3$ Atomic masses:
(Zn) = 65.38, O = 16, Ag = 108, N = 14, H = 1, N = 14, O = 16, Cl = 35.5, Ca = 40

SECTION A (16 Marks)

Answer all questions in this section

- 1 Write down the letter of the most correct answer in the answer sheet provided in each question:
 - (i) Chemist prepared hydrogen gas by mixing zinc granules with dilute hydrochloric acid. Which one of the following is correct about zinc react?

A . it is oxidized	D. it is reduced
B. it is an oxidizing agent	E. it is reducing agent
C. it undergoes both oxidation and reduction	
 - (ii) why is methane used as a source of fuels at homes and industries?

A. It burns in air to give out energy
B. it burns in air to give out water vapour
C. it burns in air to absorb energy
D. it burns in air to absorb water vapour
E. it burns in air to give out carbon dioxide.

- (iii) which of the following is a natural acid?
 A nitric acid
 B. phosphoric acid
 C. Citric acid
 D. sulfuric acid
 E. hydrochloric acid
- (iv) The name given to the force of attraction that holds atoms together to form a molecule
 A, chemical change
 B, chemical bond
 C, friction
 D, centripetal force
 E. gravitation force
- (v) In a neutral atom, which sub-atomic particles are equal in number to the protons?
 A. Electrons
 B. Neutrons
 C. Electrons of its ion
 D. Neutrons of its ion
 E. both protons and neutrons
- (vi) A form one student saw a flammable sign on a box. This suggest that the box is most likely to contain;
 A. firewood
 B. paper
 C. radioactive material
 D. spirit used in lamp
 E. maney
- (vii) your young brother was given a syrup with a prescription that it should be well shaken before use. This is because the syrup is a;
 A . clear solution
 B. salt
 C. homogeneous mixture
 D. filtrate
 E. suspension
- (viii) What is the molarity of a solution containing 1.75moles of the solution in 3.00litres of solution?
 A . 5.25M
 B. 1.71M
 C. 0.58M
 D. 1.75M
 E. 3M
- (ix) Non-metals are generally betterthan metals.
 A conductors of electricity
 B. brittle materials
 C. malleable materials
 D. conductors of heat
 E. form basic oxide
- (x) Which of the following is not among the composition of air?
 A Noble gas
 B. carbondioxide
 C. Nitrogen
 D. Hydrogen
 E. Water vapour

2. Match each item in **List A** with its correct statement in **List B**. Write the letter of the correct match in the answer sheet provided.

List A	List B
(i) Decomposion reaction	A. Spectator ions are omitted.
(ii) Displacement reaction	B. A reaction in which two or more substances combine to form a single product.
(iii) Precipitation reaction	C. Formation of ammonia.
(iv) Redox reaction	D. Formation of oxgen gas by heating potassium chlorate in the laboratory.
(v) Synthesis reaction	E. Gaining oxygen.
(vi) Ionic equation	F. Purification of metal.
	G. A metal which is liquid at room temperature.
	H. Oxidation and reduction reactions occur simultaneously.
	I. Produce salt and water.
	J. Gaining hydrogen.

SECTION B (54 marks)

Answer all questions in this section

3. Briefly explain six economic activities that depend on water.
4. (a) (i) Justify the fact that hydrogen has got both properties of group I and group VII element. Give two point.
(ii) Why hydrogen is placed in group I rather than group VII in the periodic table.
(b) Given that element L has electronic arrangement of 2;8;8;7 and another element Y has atomic number 17, place element L in its suitable period and compared the reactions of sodium metal with element L and Y.
5. (a) With example briefly explain three uses of matter.
(b) chemistry is a science that involves experiments and practical work, chemists must acquire scientific skills to successfully obtain facts and verify them. Write down scientific skills that chemistry students should apply to be able to think logically and critically.
6. (a) A certain chemist proposed the use of hydrogen gas as fuel in vehicles. suggest three advantages and three disadvantage of using the gas as fuel.
(b) why chemists prefer zinc granules and dilute hydrochloric acid in the laboratory preparation of hydrogen gas.
7. (a) Identify six steps to follow in lighting the Bunsen burner so as to produce a non-luminous flame.
(b) Give three uses of non-luminous flame and for each indicate the property of the flame that is concerned.
8. (a) A form three student need hard water for an experiment. Unfortunately, only soft water was available in the laboratory. What are the four alternative chemicals can be added to the available water so as to perform the experiment?
(b) Explain each of the following cases with support of chemical equation.
(i) Temporary hardness of water can be removed by boiling.
(ii) Permanent hardness of water can be removed by addition of washing soda.
(c,) Explain three advantage of hardness of water in daily life.

SECTION C (30 marks)

Answer **ONLY** two questions

9. Your community is switching from using firewood to a modern fuel for cooking and heating to reduce indoor air pollution and deforestation. Explain six **important** factors you would consider when choosing the most suitable alternative fuel for wide spread adoption in your community
10. A doctor prescribed antacid tablets made from magnesium hydroxide to a student who was suffering from excess stomach Acid;
(a) why did the doctor prescribe the antacid tablets?
(b) write a balanced chemical equation that took place in the student's stomach after taking the tablets.
(c) If the normal acid content in the stomach is 160.00millimolar and the stomach secretes 0.50Litre of gastric juice had 210.00millimolar hydrochloric acid.
(d) If each tablet of an antacid weighs 14mg of magnesium hydroxide, how many tablets will the student be given?
(e) If a does requires 1 tablet per every 12 hours, how many days should the student take to finish the full dose?
11. During a chemistry class, a teacher holds up a packet of common salt and states, "Beyond compounds like this are vital in numerous daily applications." Explain **six** different uses of the compounds in our everyday lives.

KILWA DC
CHEMISTRY MARKING SCHEME TERMINAL EXAMINATION 2026

SECTION A (16 Marks)

1

i	ii	iii	Iv	v	vi	vii	viii	ix	X
A	A	C	B	A	D	E	C	B	D

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2

I	Ii	Iii	iv	V	Vi
D	F	G	H	B	A

SECTION B (54 marks)

3. The economic activities that depend on water are;

Fishing: It provides livelihood to communities living near rivers, lakes and oceans. (1.5 marks)

Agriculture depends on water for irrigation, livestock and crop production. (1.5 marks)

Hydropower generation is another economic activity that relies on water. Flowing water is used to generate electricity. (1.5 marks)

Transportation and shipping use water bodies as routes for moving goods and people. Rivers, lakes and oceans facilitate trade and reduce transportation costs. (1.5 marks)

Tourism and recreation activities such as swimming and water sports depend on clean and sufficient water. These activities generate revenue and employment to the people. (1.5 marks)

Industrial processes in sectors such as manufacturing, textiles and food processing require water for cooling, cleaning and as raw-material. (1.5 marks)

4.A(i)(a) Hydrogen exhibits properties similar to group I elements because it has one electron in its outermost shell like the alkali metals, allowing it to form positive ions (H^+). (1.5 marks)

4(i)(b) Hydrogen resembles group VII elements because it can gain one electron to form a negative ion similar to halogens. (1.5 marks)

(ii) Hydrogen is placed in group I rather than group VII in the periodic table because it has a single electron in its outer shell, similar to alkali metals, allowing it to form H^+ ion. (1.5 marks)

4.B,

Element L has 25 electrons.

Element Y has atomic number 17.

-Electronic configuration of L corresponds to period 4, group 7. (1.5 marks)

Sodium reacts vigorously with both L and Y because they are halogens forming salts. (1.5 marks)

but reactions with L may be slightly less vigorous than with Y (ion) due to its atomic size and bond strength. (1.5 marks).

5.(A)Uses of matter with examples are three points:

- i)used for drink example water.(1mark)
- ii)used for cutting example knife.(1mark)
- iii)used for transportation example car.(1mark)

5.(B)Scientific skill include:

- 1.making thorough observation.(1mark)
- 2.Recording accurately what has been observed.(1mark)
- 3.organising the observed and recorded information.(1mark)
- 4.Repeating tests to ensure observations are accurate.(1mark)
- 5.Drawing conclusion from observation.(1mark)
- 6.predicting possible outcomes of similar experiments.(1mark)

6.A)Three advantages of using hydrogen gas as fue lin vehicles:

- Hydrogen produces clean energy with water as the only by-product, reducing pollution.
- It has high energy content per unit mass, providing efficient fuel.
- Hydrogen can be produced from renewable sources, making it sustainable.

Three disadvantages of using hydrogen gas as fuel:

- High cost in production and storage.
- Highly flammable, posing safety risks.
- Infrastructure for distribution is limited, hindering wide spread use.

1mark @ total 6 marks

(B)Chemists prefer zinc granules and dilute hydrochloric acid to pure zinc and concentrated acid for laboratory hydrogen preparation because granules increase surface area for the reaction. Dilute acid minimizes violent reactions, improving safety and efficiency. 3marks

7(A)Six steps to follow in lighting a Bunsen burner to produce a non-luminous flame:

- I. Connect the rubber tube to the gas tap.
- II. Close the airhole of the burner by turning the collar.
- III. Light the gas at the top of the barrel with a lighted matchstick.
- IV. Slowly turn on the gas to ignite the burner.
- V. Open the air hole gradually by rotating the collar to allow more air into the burner.
- VI. Adjust the air hole until a blue non-luminous flame is produced.

1mark @ Total 6 marks

7B.Three uses of non-luminous flame:

- Heating chemicals without soot formation.

Property: clean combustion.

- Sterilizing inoculating loops.

Property: high temperature.

•Performing flame tests formetalions.

Property:non-interfering flame color. 1mark@ total 3marks

8 (a) Four alternative chemicals to make soft water suitable for experiments.

Adding calcium chloride, magnesium sulfate, sodium carbonate, or washing soda to adjust hardness.

1mark @ total 4 marks

(b) Temporary hardness can be removed by boiling because calcium bicarbonate decomposes on heating: $\text{Ca}(\text{HCO}_3)_2 \rightarrow \text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O}$. (1.5 marks)

Permanent hardness can be removed by adding washing soda (Na_2CO_3), which reacts with calcium and magnesium ions to form insoluble carbonates: $\text{Ca}^{2+} + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3\downarrow + 2\text{Na}^+$. (1.5 marks)

(C) Three advantages of hardness of water in daily life include providing essential minerals like calcium and magnesium for health, reducing corrosion of pipes, and giving better taste to drinking water. 1mark@ total 3marks

SECTION C (30)

Question 9 (15 Marks)

Important factorsyou would consider when choosing the most suitable alternative fuel

- **Energy value (calorific value);** this is the amount of energy released by a fuel during its combustion. A good fuel should have **high energy value**. Energy value is determined by the amount of energy produced per unit mass of the fuel.
- **Velocity of combustion;** this refers to the rate at which a fuel burns. A good fuel should burn **with a moderate velocity** for a continuous supply of heat. It should not burn too fast or too slowly.
- **Ignition point;** this is the temperature to which the fuel must be heated before it starts burning. A good fuel should have **an average ignition point**. Low ignition point is risk due to fire hazards while high ignition point makes it difficult to start a fire. Fuel with high ignition point is safe to transport and store
- **Non-combustible material content;** these are materials left in form of ashes after burning a fuel.A good fuel should have **low or no contents** of noncombustible material. High content ofnon-combustible materials lowers the heat value of the fuel.
- **Poisonous products of combustion;** A good fuel should **not give off poisonous gases** during combustionand also it should **give off very little or no smoke**.
In general, the combustion of a good fuel should **not produce** harmful substances like soot and toxic substances.
- **Pyrometric burning effect;** this is the highest temperature that can be reached by the burning fuel. A good fuel should have **high Pyrometric** burning effect. Gaseous fuels have the highest pyrometric burning effect.
- **Availability;** A good fuel should be readily available in large quantities.
- **Affordability;** A good fuel should be cheap and affordable to most people.
- **Ease to transport and store;** a good fuel should be easy and safe to transport, handle and store.

- **Effect in the environment;** a good fuel should not pollute the environment during its production, storage and use. Example Fossil fuels which produce carbon monoxide on burning are major contributor to **environmental pollution**.

MARKS ALLOCATION:

Introduction: **1.5marks**

Any six factors: **2 marks @ = 12 marks** Conclusion: **1.5marks**

Question 10(15 Marks)

A The doctor prescribed antacid tablets because magnesium hydroxide neutralizes excess stomach acid, relieving pain and discomfort such as heartburn (1.5marks)

B. balanced chemical equation



C. Calculation

Normal acid = 160mM

Given acid = 210mM

Excess acid;

$$210 - 160 = 50\text{mM} \quad (1.5\text{marks})$$

$$50\text{mM} = 0.05\text{mol/L}$$

Volume of gastric juice = 0.50L

Moles of excess HCl

$$0.050 \times 0.50 = 0.025\text{mol} \quad (1.5\text{marks})$$

From the equation;

1molMg(OH)₂ reacts with 2mol of HCl

$$\text{Moles of Mg(OH)}_2 \text{ needed} = 0.025 \div 2 = 0.0125\text{mol} \quad (1\text{mark})$$

Molar mass of Mg(OH)₂ = 58.3g/mol

Mass required = 0.0125 × 58.3

$$= 0.729\text{g} \quad (3 \text{ marks})$$

D. Number of tablets

Total mass of dose = 0.729g = 729mg

Each tablet = 145mg

$$729 \div 145 = 5$$

Number of tablets is 5 (2marks)

E Duration of dose

Dose 1 tablet every 12hour thus 2 tablets per day

Thus number of tablets are 5

$$5 \div 2 = 2.5\text{days}$$

Thus duration of dose is 3 days

(3marks)

Question 11 (15 Marks)

Salts have a lot of uses in our daily lives. Some of them are explained below:

- **Control soil pH;** soils that are too acidic or alkaline are not suitable for crop production. Calcium carbonate is usually added when soil is acidic to neutralize the acidity and calcium sulphate (gypsum) is usually added when soil is alkaline to neutralize the alkalinity condition.
- **Used as antacid;** some salts are used to relieve heartburn and acid build-up in the stomach. Some common antacids are magnesium sulphate, magnesium hydrogen carbonate, and sodium hydrogen.
- **Used as inorganic fertilizer;** some salts are added in the soil as fertilizers to increase fertility. Examples of those salts are ammonium sulphate, ammonium nitrate, potassium nitrate and calcium phosphate.
- **Alleviation of health disorders;** various salts are used to alleviate problems that are associated with different health conditions. These salts include chlorides, phosphates and sulphates of sodium, potassium and calcium.
- **Production of pesticides;** some of the salts are used as pesticides to kill or control pests such as insects, rodents, weeds and fungi. For example, copper (II) sulphate, iron (II) sulphate and sodium chlorate (V) are used as pesticides for control of fungi in crops.
- **Softening of hard water;** salts such as sodium carbonate can be used to soften hard water.
- **Preservation of food;** sodium chloride or common salt due to its property of absorbing water from substances can be used in food preservation since when added to moist food it absorbs the moisture and leave the food dry.
- **Used as drying agents;** some salts are hygroscopic; hence they can be used as drying agents in different chemical processes, such salts include sodium chloride, anhydrous calcium chloride and copper (II) sulphate. For example, anhydrous calcium chloride is used as drying agent of different gases except ammonia.
- **Improve taste of food;** salts such as sodium chloride is added in food to improve the taste of the food.
- **Used in fire extinguishers;** sodium hydrogen carbonate is used as a chemical composition of dry chemical fire extinguisher which is used to put off class A, B, C and E fires.
- **Used in construction;** salts such as calcium carbonate plays an important role as a construction and building material or as an ingredient in cement.
- **Used in manufacture of dry batteries;** some salts such as ammonium chloride are used as electrolyte in manufacture of dry batteries.

- **Used in water treatment plants;** in urban water treatment plants salts are used. For example, aluminium sulphate and copper sulphate is added in water to clump together fine solid particles that are allowed to settle at the bottom of the tank.
- **MARKS ALLOCATION:**
Introduction: **1.5marks**

Any six uses of salts: **2 marks @ = 12 marks** Conclusion: **1.5marks**